$$\int_{x_0}^{x} f(x)\,\mathrm{d}x = F(x) - F(x_0)$$

$$\sum_{n=1}^{\infty} \frac{1}{n^2+1} = \frac{\pi^2}{12}$$

$$\lim_{t\to\infty} \ln t = \int_1^{\infty} \frac{1}{t}\,\mathrm{d}t$$

$$\|\boldsymbol{a}\| = \sqrt{a_1^2 + a_2^2}$$

$$\overrightarrow{PS} = \left\langle x^2, \frac{x}{2} \right\rangle$$

$$\iiint_V \nabla \cdot F(\boldsymbol{x})\,\mathrm{d}V = \oiint_S F(\boldsymbol{x}) \cdot \boldsymbol{n}\,\mathrm{d}S$$

$$\prod_{k=1}^{\infty} \left(1 + \frac{1}{k^2}\right) = \frac{\sinh \pi}{\pi}$$

$$\iint_S f(x, y)\,\mathrm{d}x\,\mathrm{d}y$$

$$\oint_C f(z)\,\mathrm{d}z$$

$$\int_1^2 x^2\,\mathrm{d}x = \left.\frac{x^3}{3}\right|_1^2$$

$$A = \bigcup_{n=1}^{\infty} A_n$$